

LATIN SQUARES OF SMALL RANK

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ABSTRACT. Let $r(n)$ be the least real rank of an $n \times n$ Latin square with entries $1, \dots, n$. On MathOverflow, Philip Weiss asked whether $r(n)$ can stay bounded as n grows, and observed that $r(2^k) \leq k+1$. We show that every Latin square of order n has rank at least $1 + 3(n-1)/(n+1)$, with strict inequality when n is odd, so $r(n) \geq 4$ for all $n \geq 5$. We also determine $r(n)$ for $n \leq 8$: the values for $n = 1, \dots, 8$ are $1, 2, 3, 3, 5, 4, 6, 4$. The lower bound and the values $r(4) = 3$, $r(6) = 4$, $r(8) = 4$ are formally verified in Lean 4; $r(5) = 5$ and $r(7) = 6$ rest on an exhaustive search with an exact certificate. Whether $r(n)$ is unbounded remains open.

1. THE QUESTION

A Latin square of order n is an $n \times n$ array in which each of the symbols $1, \dots, n$ occurs once in every row and once in every column. Read as a real matrix, it has a rank. The cyclic square, with entry $i + j \bmod n$ in row i and column j (written with values $1, \dots, n$), has rank $1 + \sum_t \varphi(p_t^{a_t})$ when $n = \prod_t p_t^{a_t}$ [4], which is about n for prime n . Other squares do much better. The square

$$V = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix}$$

has rank 3: every column is a combination of $(1, 1, 1, 1)$, $(0, 1, 0, 1)$ and $(0, 0, 1, 1)$. It is the case $k = 2$ of a construction by Philip Weiss [5]: for $n = 2^k$, put $1 + (i \oplus j)$ in row i and column j , where $\oplus$ is bitwise exclusive or, and every column lies in the span of the all-ones vector and the k bit vectors. So $r(2^k) \leq k + 1$, where

$$r(n) = \min\{\text{rank } V : V \text{ a Latin square of order } n \text{ with entries } 1, \dots, n\}.$$

Weiss asked whether $r(n)$ is bounded along some sequence of n , and whether the logarithmic bound can be improved.

Before reading on, try the next case. Can a 5×5 Latin square have rank 3? Rank 4? The answer is no to both: $r(5) = 5$, so every 5×5 Latin square is invertible. The general statement is the following.

Theorem 1. *Every Latin square V of order $n \geq 1$ satisfies*

$$\text{rank } V \geq 1 + \frac{3(n-1)}{n+1},$$

and the inequality is strict when $n \geq 3$ is odd.

Corollary 2. $r(n) \geq 4$ for every $n \geq 5$, and $r(4) = 3$.

Proof. For even $n \geq 6$ the bound exceeds 3, and for odd $n \geq 5$ it is at least 3 and strict, so in both cases $\text{rank } V \geq 4$. For $n = 4$ it gives $\text{rank } V \geq 2.8$, so $\text{rank } V \geq 3$, and the square above attains 3. $\square$

The bound tends to 4, so Theorem 1 cannot decide whether $r(n)$ is bounded. For small orders we can compute $r(n)$ exactly (Section 3).

n	1	2	3	4	5	6	7	8
$r(n)$	1	2	3	3	5	4	6	4
$1 + \log_2 n$	1	2	2.58	3	3.32	3.58	3.81	4

Table 1. The least rank of a Latin square of order n . For $n = 4$ and $n = 8$ the exclusive-or square is optimal.

2. PROOF OF THEOREM 1

Centring. Every row and column of V sums to $n(n+1)/2$. Put $C = 2V - (n+1)J$, where J is the all-ones matrix. Each row and column of C is an ordering of the n numbers

$$-(n-1), -(n-3), \dots, n-3, n-1,$$

which sum to 0 and whose squares sum to $S = n(n^2 - 1)/3$. So every row and column of C sums to 0, every column has squared length S , and every row contains the entry $n-1$.

The right subspace. Let $W \subseteq \mathbb{R}^n$ be the column space of V and $\mathbf{1}$ the all-ones vector. Since $V\mathbf{1} = \frac{n(n+1)}{2}\mathbf{1}$, the vector $\mathbf{1}$ lies in W . Each column of C is 2 times a column of V minus $(n+1)\mathbf{1}$, so it lies in W , and it is orthogonal to $\mathbf{1}$ because its entries sum to 0. So all columns of C lie in

$$K = W \cap \mathbf{1}^\perp, \quad \dim K = \text{rank } V - 1 =: d.$$

Lemma 3. *There are vectors $x_1, \dots, x_n$ and $y_1, \dots, y_n$ in $\mathbb{R}^d$ with*

$$C_{ij} = \langle x_i, y_j \rangle, \quad \sum_{i=1}^n \|x_i\|^2 = d, \quad \|y_j\|^2 = S.$$

Proof. Take an orthonormal basis $u_1, \dots, u_d$ of K , let $x_i = (u_1(i), \dots, u_d(i))$ be the i -th coordinates of the basis vectors, and let $y_j = (\langle u_1, c_j \rangle, \dots, \langle u_d, c_j \rangle)$ be the coordinates of the j -th column c_j of C in this basis. Expanding $c_j = \sum_k \langle u_k, c_j \rangle u_k$ and reading off entry i gives $C_{ij} = \langle x_i, y_j \rangle$. The sum $\sum_i \|x_i\|^2$ is $\sum_k \|u_k\|^2 = d$, and $\|y_j\|^2 = \|c_j\|^2 = S$ because the basis is orthonormal. $\square$

In words: a Latin square of rank $d+1$ is the same thing as n row points and n column points in $\mathbb{R}^d$ whose inner products are the centred entries. Figure 1 shows the square above this way, with $d = 2$.

Proof of Theorem 1. Row i contains $n-1$, say in column j . By Cauchy–Schwarz,

$$(n-1)^2 = \langle x_i, y_j \rangle^2 \leq \|x_i\|^2 \|y_j\|^2 = \|x_i\|^2 S.$$

Adding over the n rows and using $\sum_i \|x_i\|^2 = d$ gives $n(n-1)^2 \leq dS = dn(n-1)(n+1)/3$, that is, $3(n-1) \leq (n+1)d$. This is the inequality, since $d = \text{rank } V - 1$.

Suppose now that $n \geq 3$ is odd and $3(n-1) = (n+1)d$. Then the sum of the n inequalities is an equality, so each one is: $\|x_i\|^2 S = (n-1)^2$ for every i . Fix a row i and let $n-1$ and $-(n-1)$ sit in columns j and j' . Then $\langle x_i, y_j \rangle = \|x_i\| \|y_j\|$ and $\langle x_i, -y_{j'} \rangle = \|x_i\| \|y_{j'}\|$, and equality in Cauchy–Schwarz, with $\|y_j\| = \|y_{j'}\|$ and $x_i \neq 0$, forces $y_{j'} = -y_j$. Then $C_{kj'} = -C_{kj}$ for every k : column j' of C is minus column j . Since n is odd, 0 is one of the centred values, so column j has a 0 in some row k , and so does column j' . Row k then contains 0 twice, in columns $j \neq j'$, which is impossible. $\square$

For even n the centred entries are odd, there is no 0 to use, and equality is possible in principle; it would need $(n+1) \mid 3(n-1)$, which happens only for $n = 2$.

3. EXACT VALUES

Rank does not change when rows or columns are permuted. So every Latin square of order n has the rank of one obtained from a representative of its isotopy class, with the n symbols given the values $1, \dots, n$ in some order. McKay lists one representative of each class [2]: there are 2,

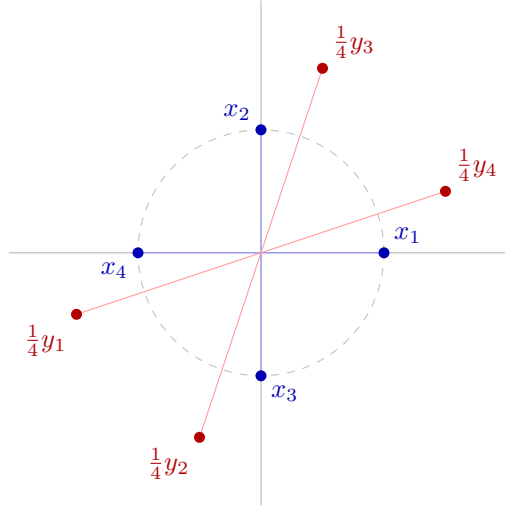


Figure 1. The 4×4 square of rank 3 as points in the plane. Its centred form C has rows $(-3, -1, 1, 3)$, $(-1, -3, 3, 1)$, $(1, 3, -3, -1)$, $(3, 1, -1, -3)$, and $C_{ij} = \langle x_i, y_j \rangle$. The row points lie on the circle of radius $1/\sqrt{2}$, so $\sum \|x_i\|^2 = 2$; the column points (drawn at a quarter of their length) have $\|y_j\|^2 = 20$. The entry 3 in each row needs $9 \leq \|x_i\|^2 \cdot 20 = 10$.

2, 22 and 564 classes for $n = 4, 5, 6, 7$ [3]. We computed the rank of all $(\# \text{classes}) \times n!$ resulting matrices modulo the prime $p = 2^{31} - 1$. The rank modulo p never exceeds the rank over $\mathbb{Q}$, so the minimum over all of them is an exact lower bound for $r(n)$, and a matrix attaining it, whose rank over $\mathbb{Q}$ we computed exactly, shows the bound is sharp. This gives $r(4) = 3$, $r(5) = 5$, $r(6) = 4$ and $r(7) = 6$. The values $r(2) = 2$ and $r(3) = 3$ follow from Theorem 1, whose bound is 2 for $n = 2$ and exceeds 2.5 for $n = 3$, and $r(1) = 1$.

For $n = 6$ and $n = 8$ Corollary 2 gives $r(n) \geq 4$, and rank-4 squares exist. For $n = 8$ the exclusive-or square works. For $n = 6$ one example is

$$\begin{pmatrix} 1 & 6 & 2 & 5 & 3 & 4 \\ 6 & 1 & 5 & 2 & 4 & 3 \\ 2 & 5 & 3 & 4 & 1 & 6 \\ 5 & 2 & 4 & 3 & 6 & 1 \\ 3 & 4 & 6 & 1 & 5 & 2 \\ 4 & 3 & 1 & 6 & 2 & 5 \end{pmatrix},$$

which is AB for an integer 6×4 matrix A and 4×6 matrix B (listed in the Lean files). So $r(6) = r(8) = 4$, without any search.

4. REMARKS

The argument of Theorem 1 uses only two facts: each row contains the extreme value $n - 1$, and the columns have equal length. It cannot give more than 4, since $3(n - 1)/(n + 1) < 3$. Table 1 is consistent with $r(n) \geq 1 + \log_2 n$, which would make the exclusive-or square optimal at every power of 2 and would show that $r(n)$ is unbounded. We have no proof.

The orders 5 and 7 are striking: no Latin square of order 5 is singular, and none of order 7 has rank below 6.

5. VERIFICATION

Lemma 3, Theorem 1 with its strict form, Corollary 2, and the values $r(4) = 3$, $r(6) = 4$, $r(8) = 4$ are proved in Lean 4 with Mathlib [1], using only the standard axioms. The low-rank squares are checked there to be Latin and to factor as AB over $\mathbb{Z}$. The values $r(5) = 5$ and $r(7) = 6$ rest on the computation of Section 3, which checks every class representative, uses

exact modular arithmetic, and is caught by a planted error (ranks computed modulo 3). Code, data and Lean files are at [6].

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